

인공지능과 뇌과학(이) (BR407-01)

학기	2026년 2학기
담당교수	김민환
이수구분	교선평수
학점	3.0
강의시간	화16:30-18:00(E7 - L28), 목16:30-18:00(E7 - L28)
강의실	E4 411 (교수실)
소속	기초학부

교과개요

This course explores the relationship between biological and artificial intelligence through direct comparison. In the first half of the semester, students study five major comparison components: representation, learning rules, credit assignment, action selection and control, and efficiency/hardware. In the second half, students gain hands-on experience with spiking neural networks using snnTorch. In the final part, students critically evaluate review literature in neuroscience-inspired AI through solo or group presentations.

교과목표

- Explain major computational principles in biological and artificial intelligence
- Compare biological and artificial representations of information
- Distinguish local biological plasticity rules from AI optimization methods
- Describe spatial and temporal credit assignment problems
- Explain reinforcement learning concepts in relation to dopamine and basal ganglia circuits
- Implement basic spiking neural network models using snnTorch
- Critically analyze review literature in neuroscience-inspired AI
- Evaluate the strengths and limits of neuroscience-AI analogies

수업방법

- 1주차 : Course Introduction: Why compare Neuroscience and AI?
- 2주차 : Component 1: Representation
- 3주차 : Component 2: Learning Rules
- 4주차 : Component 3: Credit Assignment
- 5주차 : Component 4: Action Selection and Control
- 6주차 : Component 5: Efficiency and Hardware
- 7주차 : Integration Week (Review): Bridging the Five Components
- 8주차 : Mid-term
- 9주차 : Practical Bridge: Introduction to snnTorch
- 10주차 : snnTorch Practical I: Spike Encoding and LIF Neurons
- 11주차 : snnTorch Practical II: Feedforward Spiking Neural Networks
- 12주차 : snnTorch Practical III: Training SNNs
- 13주차 : snnTorch Practical VI: Surrogate Gradient Descent in a Convolutional SNN
- 14주차 : snnTorch Practical V: Neuromorphic Datasets with Tonic + snnTorch
- 15주차 : Review Paper: Presentation I
- 16주차 : Review Paper: Presentation II

학습윤리

Students must complete all course work honestly in writing, coding, and presentations. Any use of outside sources, including papers, websites, code, or AI-generated content, must be clearly acknowledged. Plagiarism, unauthorized collaboration, fabricated results, or copied code without proper citation will be treated according to university academic integrity policy.

교과정책(방침)

Letter grade (ABCDF)

준비물 및 기타

- Kandel et al., Principles of Neural Science (Fifth Edition)
- Sutton and Barto, Reinforcement Learning (Second Edition)
- <https://snntorch.readthedocs.io/en/latest/index.html>